

Mode 3 — Baseline-BFP

DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → block exponents

Step 1 — DP8 (16×8)

$$\text{DP8 (16} \times 8) = \sum_{i=0}^7 a_i b_i$$

Step 2 — DP8 (16×4)

$$\text{DP8 (16} \times 8) = 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 (16} \times 4)} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 (16} \times 4)}$$

Step 3 — DP8 (8×4)

$$\text{DP8 (16} \times 8) = 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hi}}_{\text{DP8 (8} \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lo}}_{\text{DP8 (8} \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hi}}_{\text{DP8 (8} \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lo}}_{\text{DP8 (8} \times 4)}$$

Step 4 — Block exponents (one exponent block per DP8 (8×4) — all copies of one E)

$$(P_j, E) \text{ per DP8 (8} \times 4) \text{ above,} \quad E = E_a + E_b \longrightarrow (M, E)$$

$\delta \equiv 0$: M is the Step-3 sum unchanged; $\text{DP8 (16} \times 8) = M \cdot 2^E$.

Step 5 — Result

$$M = 2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{hi} + 2^8 \sum_{i=0}^7 a_i^{hi} b_i^{lo} + 2^4 \sum_{i=0}^7 a_i^{lo} b_i^{hi} + \sum_{i=0}^7 a_i^{lo} b_i^{lo}, \quad E = E_a + E_b$$